# MAASAI MARA UNIVERSITY

## REGULAR UNIVERSITY EXAMINATIONS

## 2022/2023 ACADEMIC YEAR

## SECOND YEAR SECOND SEMESTER

## SCHOOL OF PURE, APPLIED AND HEALTH SCIENCES

## BACHELOR OF SCIENCE IN COMPUTER SCIENCE

## COURSE CODE: COM 2216-I

## COURSE TITLE: WEB DESIGN AND DEVELOPMENT

**DATE: 21/4/ 2023** **TIME: 0830-1030 HRS**

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**INSTRUCTIONS TO CANDIDATES**

- ANSWER Question **ONE** and any other **TWO**

## QUESTION ONE (20 MARKS)

a.) Explain five key distinct differences between a web designer and a web developer. **[5 marks]**

b.) Jane is a web development project manager who began site development after all stakeholders signed off on the project plan. However, over the course of the project, several stakeholders continuously asked Jane to add additional sections and functionality to the site. These requests appeared small and seemed to have only a minimal impact on the development team when they occurred, but Jane realized that, considered together, these additional tasks would increase the cost and time to complete the project to an unacceptable degree. Explain what this uncontrolled expansion is, what it can lead to, and how it can be adequately managed so as not to compromise the success of her project.

**[5 marks]**

c.) Explain the meaning of each line while showing the output of the following code snippet.

```
$str = "Hello Exam";

echo strrev($str);
```

**[2 marks]**

d.) Write PHP code that defines a function called "person_Profile" that takes in three parameters: $name, $city, and $job. It then uses the "echo" function to output a sentence that includes the values of those parameters. The function is supposed to be called three times with different arguments, and the output to be concatenated with empty strings in between. The final output should be three separate sentences, each containing the name, city, and job of a different person. **[4 marks]**

e.) What is the difference between client-side and server-side scripting, and which one is better for web development? **[4 marks]**

## QUESTION TWO (15 MARKS)

a.) What are cookies? How are cookies created and destroyed?

**[4 marks]**

b.) Passwords must never be stored in the database as plain text. Discuss three ways that password storage can be handled for web applications so that they are not only safe, but also strong and not easily decryptable from pre-computed tables. Use simple examples. **[7 marks]**

c.) You are asked to use arrays to store a person's profile data. What do you understand by an array. Explain what type of array this code represents and state where its suitable for use:

```
$arr = array("name" => "John", "age" => 20, "city" => "New York")
```

**[4 marks]**

## QUESTION THREE (15 MARKS)

a.) As a web developer explain how you can optimize your site for search engine ranking using at least six best practices for SEO in web development. **[6 marks]**

b.) Responsive web design is an approach to web design that aims to create a website that looks great and works well on all devices, including desktops, laptops, tablets, and mobile phones. Explain five key principles of responsive web design. **[10 marks]**

c.) Distinguish between the following array operators

*$x == $y* versus *$x === $y*

*$x <> $y* versus *$x !== $y*

**[4 marks]**

## QUESTION FOUR (15 MARKS)

a.) What are sessions used for? Show the syntax and explain why they are important. **[3 marks]**

b.) You are asked to develop a web-based login page for user authentication to the National Defence Data Center (NDDC) database. You are expected to write code for the NDDC's web page interface containing form input for username and password and initiate the user session. Secondly write a complete connection string to retrieve the user's data from the database for authenticating this particular user. Users that are allowed to the NDDC page should be ushered into welcome.php page. Use appropriate error messages incase of user authentication failure.

**[12 marks]**

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